

Module 01 - Assignment

Q1. What is Software Quality Assurance (SQA)?

Define SQA:

A planned, systematic set of activities to ensure software conforms to established quality standards is called Software Quality Assurance. It's not about the product per se but the process. The idea is that you don't catch the defect at the end you prevent it before it can even occur. SQA can be used throughout the life cycle (needs, development, production support).

Key Points:

- Quality = Meeting customer expectations
- Assurance = Preventing defects/bugs

Why quality is important in software development:

1. **Trust & Reputation:** One big mistake (such as payment failure) can cause customer trust to be violated in mere moments.
2. **Revenue Protection:** With ecommerce or banking apps, no time lost means no money made, and bugs lead to lost time which means lost money.
3. **Security & Compliance:** In the regulated sector, such as banking and health care, there are legal sanctions for failing to meet standards.
4. **User Retention:** The first reason after one experience when Users quit an app is because of poor quality.
5. **Long-Term Cost:** Quality debt - Things that can be fixed yesterday, cost quite a bit more to fix today.

Q2. Differentiate between the following:

QA vs QC

Aspect	Quality Assurance (QA)	Quality Control (QC)
Focus	Process-oriented	Product-oriented
Goal	Prevent defects	Identify defects
Approach	Proactive	Reactive
Activities	Audits, standards, process reviews	Inspection, reviews of the product
Responsibility	Entire team	QC / QA team
Example	Defining a code review checklist	Reviewing a build against that checklist

QC vs Testing

Aspect	Quality Control (QC)	Testing
Focus	Product-oriented	Execution-oriented
Goal	Identify defects	Find bugs & verify behavior
Approach	Reactive	Reactive
Activities	Inspection, reviews of the product	Test case execution, exploratory testing
Responsibility	QC / QA team	Testers / Automation Engineers
Example	Reviewing a build against a checklist	Running login test cases on that build

QA vs Testing

Aspect	Quality Assurance (QA)	Testing
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Q3. How is AI changing Software Testing?

AI is revolutionizing the software testing sector in several important ways:

- 1. From Scripted to Intelligent:** When talking about traditional automation, you are listening to something which works on set scripts. AI-driven testing adapts to change and learns from past runs, and understands intent.
- 2. Self-Healing Automation:** In case of UI-change in a locator, it is automatically re-generated and once new locators are identified by AI, the test will be automatically updated accordingly effort level of maintenance can be substantially reduced.
- 3. AI Across the SDLC:** While execution is not the only use of AI for requirements, requirement analysis, test case generation, code review, execution and even defect triage are all now supported by AI.
- 4. Intelligent Reporting:** AI parses failures, groups together related failures and asks you to focus on root causes rather than just parsing failure passes/fails.
- 5. AI-Powered Testing Tools:** AI-powered features such as Claude Code, GitHub Copilot, Gemini and ChatGPT are being employed to support a variety of testing functions beyond code-testing.

Q4. Explain the responsibilities of:

QA Engineer:

QA Engineers create extensive test cases, data, automation scripts, know how to analyze bugs, and test AI-enabled features.

Software Test Engineer:

Testers and Automation Engineers have the responsibility for testing. They want to discover bugs and runs tests for behavior, taking a reactive approach.

AI Quality Engineer:

The AI Quality Engineer is based on accuracy, hallucination, context, prompt understanding and security. Three key skills needed are: Traditional QA, AI Prompting, and Coding.

Daily Activities

QA Engineer:

1. Requirement Analysis
2. Test Case Writing
3. Test Data Creation
4. Test Execution
5. Bug Report Writing
6. Automation Coding
7. Documentation

Software Test Engineer

1. Execution of tests and exploratory testing.

AI Quality Engineer

1. AI Assisted Requirement Analysis
2. AI Generated Test Cases
3. Generate test data using AI.
4. Create test data with AI.
5. AI Assisted Test Execution
6. Automated bug reporting with the help of AI.
7. AI Assisted Automation Script Generation
8. AI Assisted Documentation

Q5. Draw the career roadmap of a QA professional.

